April/16

Firm : R recruiters
P producers

$$R + P = N_{workers}$$

$$\boxed{\tau = \frac{R}{P}}$$ τ (Recruiter-Producer ratio)

workers lost

$$V = \frac{s \cdot N}{q(\theta)}$$

not all vacancies are filled at every moment

→ new hires

$$= \underbrace{\frac{s \cdot N}{q(\theta)}}_{V} \cdot \underbrace{q(\theta)}_{\text{probability of finding workers}} = \underbrace{s \cdot N}_{\text{workers to be replaced}}$$

$$R = r \cdot V = \frac{r \cdot s}{q(\theta)} \cdot N$$

$$R = \frac{r \cdot s}{q(\theta)} \cdot (R + P)$$

$$R\left(1 - \frac{r \cdot s}{q(\theta)}\right) = \frac{r \cdot s}{q(\theta)} \cdot P$$

$$\tau = \frac{R}{P} = \frac{\frac{r \cdot s}{q(\theta)}}{1 - \frac{r \cdot s}{q(\theta)}} \quad \therefore \quad \boxed{\tau(\theta) = \frac{r \cdot s}{q(\theta) - r \cdot s}}$$

$$L^s(\theta) = \frac{f(\theta)}{s + f(\theta)} \cdot H$$

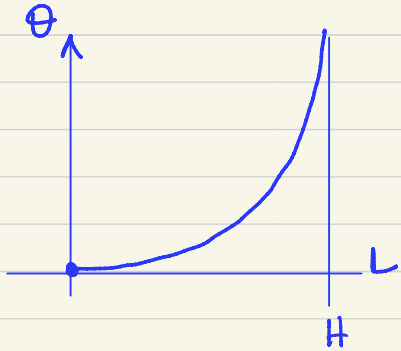
↳ size of the labor force

$f(\theta)$: job-finding rate

$$m(U,V) = \omega \cdot U^{\eta} \cdot V^{1-\eta}$$

$$f(\theta) = \frac{m}{U} = \omega \cdot \theta^{1-\eta}$$

then, $$L^s(\theta) = \frac{\omega \cdot \theta^{1-\eta}}{s + \omega \cdot \theta^{1-\eta}} \cdot H$$



$L^s(\theta = 0) = 0$

$L^s(\theta \to \infty) = H$

↳ DD: $$\frac{\omega}{\frac{s}{\theta^{1-\eta}} + \omega} \to 1$$